

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.3: The Periodic Table — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: The Periodic Table
Driving Question	DRIVING QUESTION: How does electron arrangement influence the arrangement of elements in the periodic table, and how does it help us predict the ions and compounds elements form?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Sorting the Element Cards and Launching the DQB	We sorted the first 20 Element Cards (H to Ca) by properties and by mass. We saw that Li, Na and K, with very different masses (7, 23, 39), all react with water, with the reaction getting stronger from Li to K, while Ne and Ar do not react at all. We noted that chlorine is used to treat water in Nairobi and Mombasa, and that Ar (A_r 40) is heavier than K (A_r 39) although it comes first.	Elements can be sorted by properties and by mass into patterns, but mass alone does not explain why some elements of very different mass behave alike while neighbouring elements behave very differently. Mendeleev and Moseley faced this puzzle, and something hidden inside atoms must decide where an element belongs and how it behaves.	Our first ideas: a hidden property of atoms, possibly linked to electrons, may explain why Li, Na and K are alike and why Ne and Ar are unreactive. We do not yet know what it is. We started the DQB with the driving question and our own questions, and drew the first version of the Class Model, the 'Electron Arrangement Model', to be revised in each lesson.
Lesson 2 From Mendeleev's Gaps to Moseley's Atomic Number: Building Groups and Periods	When the first 20 Element Cards were arranged by atomic number into 4 periods (rows) and groups (columns), Li, Na and K fell in one column and He, Ne and Ar in another. Ar (Z18, A_r 40) had to be placed before K (Z19, A_r 39), and Ca (A_r 40) had the same A_r as Ar but sat in a different group.	Mendeleev (1869) ordered elements by atomic mass and left gaps for undiscovered elements. Moseley (1913) showed that atomic number is the true basis of order, and the modern table uses atomic number. A group is a vertical column and a period is a horizontal row.	Mass ordering fails for Ar and K, but atomic number gives a consistent order. The table places elements that behave alike, such as Li, Na and K, in the same group even though their masses differ. The hidden property that causes these patterns is still to be explained and will be explored in Lesson 3.
Lesson 3 Writing Electron Arrangements: Reading Group and Period from s and p Notation	We built the electron arrangements of H to Ca using counters and wrote them in energy-level form (e.g. Na 2,8,1) and s and p notation (e.g. Na 1s ² 2s ² 2p ⁶ 3s ¹). We noticed that Li (2,1), Na (2,8,1) and K (2,8,8,1) each have 1 outer electron, and	Group number is decided by the number of outer electrons (group 1 and 2 = number of outer electrons; groups 13 to 18 = outer electrons + 10, with He, 1s ² , in group 18). Period number equals the number of occupied energy levels. Elements in the	Li, Na and K sit in the same column despite masses of 7, 23 and 39 because each atom has 1 outer electron, while neighbours such as Na and Mg differ because their outer electron counts differ (1 and 2). Electron arrangement, the hidden property that

	that Ne (2,8) and Ar (2,8,8) have full outer levels.	same group have the same outer electron count even though their masses differ.	follows from atomic number, therefore decides where an element is placed in the periodic table. We still need to explain why outer electrons control how elements react.
Lesson 4 Chemical Families and Reactivity Patterns: Why Li, Na and K Behave Alike	Li fizzes steadily and floats; Na melts into a ball, darts and fizzes fast; K melts, burns with a lilac flame and is very vigorous. Ca reacts faster than Mg. Ne and Ar do not react. Cl and F compounds are used in water treatment in Nairobi and Mombasa. Fe and Cu form more than one ion.	Elements fall into families: alkali metals (group 1, 1 outer electron), alkaline earth metals (group 2, 2 outer electrons), halogens (group 17, 7 outer electrons), noble gases (group 18, full outer level) and transition elements (between group 2 and 13 from period 4). Reactivity of alkali metals increases down the group: $\text{Li} < \text{Na} < \text{K}$.	Elements in the same family behave alike because they have the same number of outer electrons, not the same mass (Li 7, Na 23 and K 39 all have 1 outer electron). Neighbours such as Na and Mg or Na and Ne differ because their outer electron counts differ. Why a full outer level makes Ne and Ar unreactive is still to be explained in Lesson 5.
Lesson 5 Stability of Atoms: Why Atoms Lose or Gain Electrons	Ne (2,8), Ar (2,8,8) and He (1s2) have full outer levels and do not react. Na (1 outer electron) and Mg (2) can reach a full outer level by losing few electrons; O (6) and Cl (7) can reach it by gaining few electrons. After moving electrons, each atom had the same arrangement as a noble gas.	A full outer level (octet of 8, or duplet of 2 for He) is a stable arrangement. Atoms with few outer electrons tend to lose them, and atoms with nearly full outer levels tend to gain electrons, so they become stable like noble gases. Atoms in the same group, such as Li, Na and K, behave alike because they have the same outer electron count.	Ne and Ar do not react because their outer levels are already full, whereas Na and K react vigorously because losing one outer electron gives a stable arrangement, and chlorine used in Nairobi and Mombasa water treatment reacts because it needs just one more electron to reach the Ar arrangement. This helps explain why elements in the same group behave alike and why neighbours such as Na, Mg, Ne and Ar differ.
Lesson 6 Ion Formation: Predicting Cations and Anions and Drawing Dot-and-Cross Diagrams	Atoms of Na, Mg, Al, K and Ca lose electrons and atoms of O, F and Cl gain electrons. The ions formed have the same electron arrangement as a noble gas: Na^+ , Mg^{2+} , O^{2-} and F^- match neon (1s2 2s2 2p6), while Cl^- , K^+ and Ca^{2+} match argon (1s2 2s2 2p6 3s2 3p6). The number of protons stays the same.	Metals with 1 to 3 outer electrons lose them to form cations; non-metals with 5 to 7 outer electrons gain electrons to form anions. The charge equals the number of electrons lost or gained, and the group number helps predict it. Dot-and-cross diagrams show the transfer of electrons, with the ions in brackets with their charges.	Sodium reacts vigorously because it easily loses its single outer electron to reach the stable neon arrangement, while neon does not react because its outer level is already full. Chlorine gains one electron to reach the argon arrangement, forming Cl^- . Elements in the same group form ions of the same charge because they have the same number of outer electrons, which helps us predict ions of elements we have never worked with.
Lesson 7 Valency and Oxidation Numbers: What Do Electron Arrangements Tell Us About Charge?	We observed that Na, Mg and Al lose 1, 2 and 3 electrons to give oxidation numbers +1, +2 and +3, while Cl gains 1 and O gains 2 to give -1 and -2. We observed that iron forms Fe^{2+} and Fe^{3+} (in FeCl_2 and Fe_2O_3) and copper forms Cu^+ and Cu^{2+} , so these transition elements have more than one oxidation number.	We learned that valency is the number of electrons an atom loses, gains or shares (a positive whole number), and the oxidation number is the same number with a sign (+ for loss, - for gain). We learned that outer electrons let us infer these numbers for main-group elements, but transition elements such as iron and copper can have variable oxidation numbers.	Elements in the same group, such as Li, Na and K, have the same outer electron count, so they all have valency 1 and oxidation number +1 and react alike despite masses of 7, 23 and 39. Atoms lose or gain electrons to reach the stable noble gas arrangement, which is why Na forms +1 and Cl forms -1. Iron and copper show that the outer-electron rule has limits, and this

			prepares us to use valencies and oxidation numbers to derive formulae next lesson.
Lesson 8 Radicals and Deriving Formulae: Predicting the Compounds Elements Form	We observed that compounds are neutral, and that ions combine in ratios that balance their charges: NaCl (1:1), MgO (1:1), CaCl ₂ (1:2), Al ₂ O ₃ (2:3), KNO ₃ and Ca(NO ₃) ₂ with radicals, and Fe ₂ O ₃ , FeCl ₂ and CuO for elements with variable oxidation numbers.	We learned that a radical is a charged group of atoms that stays together (NO ₃ ⁻ , OH ⁻ , SO ₄ ²⁻ , CO ₃ ²⁻ , NH ₄ ⁺ , PO ₄ ³⁻), and that the criss-cross method, with brackets around radicals when a subscript is needed, helps us derive formulae by swapping valencies and checking that the total positive charge equals the total negative charge.	We can explain that electron arrangement decides the ion charge (for example, the 1 outer electron of K gives K ⁺ , and 3 outer electrons of Al give Al ³⁺). The ion charges then decide how many of each ion are needed to make a neutral compound, so we can predict formulae such as NaCl, CaCl ₂ , Al ₂ O ₃ and KNO ₃ and explain why Li, Na and K form similar compounds despite their different masses.
Lesson 9 Writing and Balancing Chemical Equations: What Happens When Sodium Meets Water?	We observed that sodium reacts with water by melting into a ball, darting and fizzing, that potassium reacts even more vigorously with a lilac flame and lithium more gently, that a gas (hydrogen) forms and the solution turns red litmus blue. We also saw that the unbalanced equation Na + H ₂ O → NaOH + H ₂ has unequal numbers of hydrogen atoms on each side.	We learned that a chemical equation must have the same number of each kind of atom on both sides, and we balance it by changing coefficients only, never subscripts. We learned to use state symbols (s), (l), (g), (aq) and to write correct formulae first using valencies from Lesson 8.	Li, Na and K have very different masses but each has one outer electron, so each loses it to form a 1+ ion and follows the same equation pattern, 2M + 2H ₂ O → 2MOH + H ₂ (for example 2Na + 2H ₂ O → 2NaOH + H ₂). The pattern is decided by the outer electron count, which sets the group, and not by mass. Balanced equations for 2Na + Cl ₂ → 2NaCl, 4Al + 3O ₂ → 2Al ₂ O ₃ and Ca + 2HCl → CaCl ₂ + H ₂ follow from the ions each element forms.
Lesson 10 Final Model Revision and Presentation: Using Electron Arrangement to Explain the Periodic Table and Predict Strontium	We observed that elements in the same group (Li, Na, K) react alike despite very different masses (7, 23, 39), neighbours such as Na, Mg, Ne and Ar behave differently, and Ne and Ar do not react. We used the first 20 elements' electron arrangements and ion data to see that outer electron count matches group and noble gas arrangements match ion charges. For strontium (Z38, 2,8,18,8,2) we saw two outer electrons and five energy levels.	We learned that elements are ordered by atomic number (Moseley, 1913), that outer electrons decide the group and occupied energy levels decide the period, and that atoms lose or gain electrons to reach a stable noble gas arrangement. This lets us predict ions, valencies, oxidation numbers and formulae such as NaCl, CaCl ₂ , Al ₂ O ₃ and KNO ₃ . We also learned that sharing of electrons is explained in bonding (Sub-Strand 1.4).	The hidden property is electron arrangement. Sr has 2 outer electrons and 5 occupied levels, so it is in group 2, period 5. It loses 2 electrons to form Sr ²⁺ with a noble gas arrangement, like Mg ²⁺ and Ca ²⁺ , so its compounds are SrCl ₂ and SrO. The same reasoning explains why Li, Na and K react alike, why Ne and Ar are unreactive, and why chlorine forms Cl ⁻ for water treatment in Nairobi and Mombasa.

ATTRIBUTION AND LICENSING

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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